

Software Documentation

Software Documentation

- Software documentation in testing is written information that helps testers understand what to test, how to test and what correct behavior should be.

Importance of Software Documentation in Testing

Requirements documentation	Defines what system should do Used to create test cases Used to check expected results	E.g. "The system shall allow users log in using username and password"
Design documentation	Helps testers understand system architecture and modules to test	E.g. knowing how the reporting module connects to the database helps in data flow.
Test documentation	Created to guide testing activities and track results	E.g. test plan, test strategy, test case, defect reports, test reports
User documentation	Used to compare real system with user expectations during UAT	E.g. User manual says "Click submit to save income entry"
Maintenance documentation	Used when system is updated or fixed to support regression testing	

Test Plan

- Document that explains the overall approach for testing the system.
- Includes objectives, scope, resources, schedule, test strategy and risks.
- Example:
 - To test an Automated FMS
 - Will test login, transactions and balances
 - Testing will take 3 weeks
 - Tool used: Selenium
 - High-risk areas like money transfer will be prioritized

Testing Strategy

- A testing strategy is a high-level plan that defines how testing will be carried out in a project.
- It includes methods, levels, types, techniques, tools and processes for testing.

Components of a Testing Strategy

Scope	What will and will not be tested
Testing types	What features and aspects will be tested? e.g. functional, security tests, usability test,...
Testing levels	Unit, integration, system, UAT
Testing methods	How will testing be done? (Manual or Automated)
Test environment	Hardware and software needed (PC, databases, network, application under testing)
Tools	Automation tools, defect tracking tools etc.
Roles and responsibilities	Who does what without specific names e.g. developer fixes defects
Entry and Exit criteria	Conditions for before and after testing e.g. system is ready, approved test plan, test cases, user accounts available
Risk management	Identifies possible risks and solutions e.g. risk: limited time for testing mitigation: parallel testing/automated testing/apply risk based strategy

Types of Testing Strategy

- **1. Risk-based:** Testing focusses on high risk areas e.g. test transactions and security more than UI
- **2. Model-Based:** Focusses on models of the system e.g. dataflow – test how school fees payments move from mobile money account to school account
- **3. Methodical:** Testing follows a predefined checklist of tests – Always test:
 - Login
 - Input validation
 - Error messages

Types of Testing Strategy

- **4. Standard-based:** Testing follows industry standards such as IEEE and organizational QA processes to ensure software quality
- **5. Reactive:** Testing is done after system is developed and defects are fixed as they appear
- **6. Regression-Averse:** Focusses on re-testing after changes to ensure functionality does not break

Types of Testing Strategy

- **7. Exploratory testing:** Based on tester's experience. Does not use predefined test cases
- **8. Hybrid:** Involves a combination of two or more strategies

Test Case

- Detailed step-by-step instructions to test a scenario.
- Includes test case ID, test scenario, steps, test data, expected result.

Defect Report

- It is a formal document used to record, track and communicate issues found in a software system during testing.
- Ensures defects are not forgotten

Components of a Defect Report

Component	Description	Example
Defect ID	Unique identifier	DEF-001
Summary	Short description of defect	Login fails with valid credentials
Description	Detailed explanation	User cannot log in even with correct username and password
Steps to reproduce	How to trigger the defect	1. Open login page 2. Enter valid username 3. Enter correct password 4. Click Login
Expected result	What should happen	User logs in successfully
Actual result	What actually happens	Error message displayed
Severity	Impact of defect	Critical/Major/Minor
Priority	Urgency of fixing	High/Medium/Low
Status	Current state of defect	New/ Assigned/ Open/ In progress/ Fixed/ Retest/ Closed/ Reopened/ Duplicate, cannot reproduce etc.
Assigned to	Person responsible	Developer

Testing Report

- A testing report is a document prepared after testing that summarizes what was tested, what was discovered and the overall quality of the system.
- It communicates results, provides evidence for testing and supports decision making.

Components of a Test Report

Component	Description	Example
Test summary	What was tested	FMS
Test results	Number of test cases executed, how many passed or failed	Total test cases – 100 Passed test cases – 85 Failed test cases - 15
Defect summary	Total defects found	Critical – 3 Major – 2 Minor - 1
Test coverage	Measures how much of the application is tested	Items tested/total items x 100% 95% of system tested What does 5% mean?
Issues	Problems encountered during testing	Tester server is unstable/missing requirement/limited time for testing/no automation tools
Recommendations	Suggestions based on testing results	Do not release until critical defects are fixed
Final status	Decision on whether or not to release system	Go/No-Go



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